Lab 4

- ## Microservices benefits

  Each service is independent and can be deployed or scaled separately.

- ## Complexity introduced

  Requires inter-service communication, network handling, and failure management.

- ## Failure impact

  If product-service fails, order-service cannot complete requests.

- ## 12-factor principles

  Config via environment variables, stateless services, and isolated dependencies.

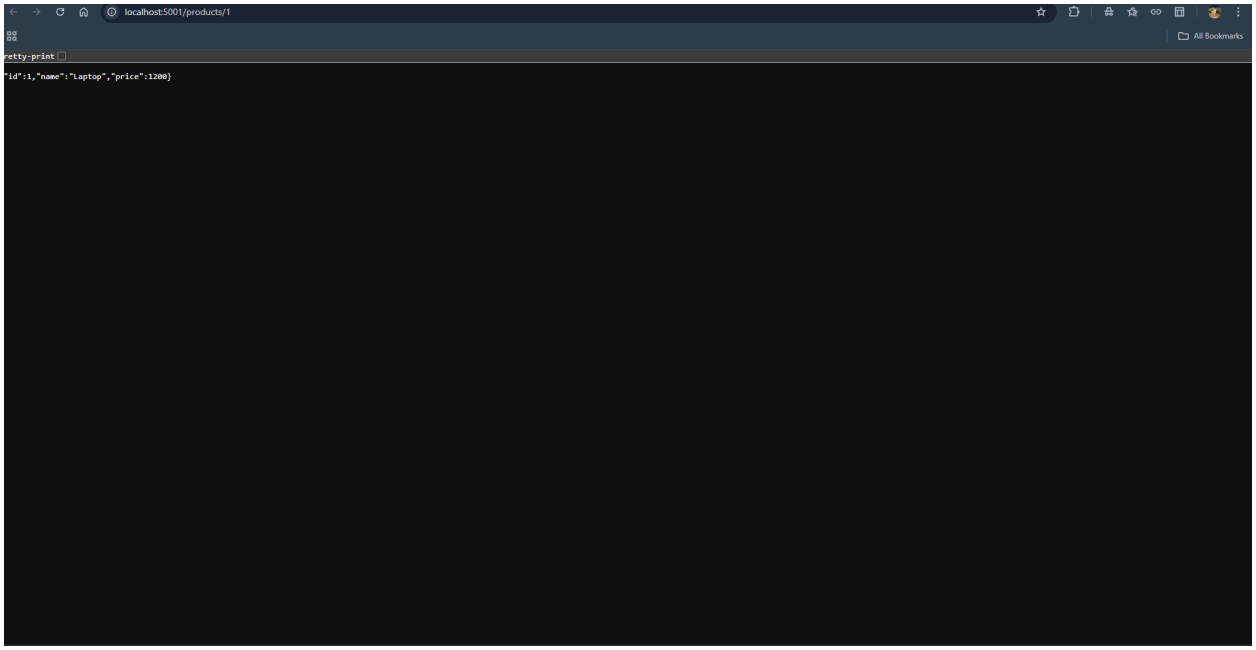
```
C:\WINDOWS\system32\cmd.

 => [product-service internal] load .dockerignore                                                    0.1s
 => => transferring context: 2B                                                                      0.0s
 => [order-service 1/5] FROM docker.io/library/python:3.12-slim@sha256:46cb7cc2877e60fbd5e21a9a      0.3s
 => => resolve docker.io/library/python:3.12-slim@sha256:46cb7cc2877e60fbd5e21a9ae6115c30ace7a0      0.1s
 => [product-service internal] load build context                                                    0.2s
 => => transferring context: 733B                                                                    0.0s
 => CACHED [order-service 2/5] WORKDIR /app                                                          0.0s
 => [product-service 3/5] COPY requirements.txt .                                                    0.1s
 => [product-service 4/5] RUN pip install --no-cache-dir -r requirements.txt                        21.6s
 => [product-service 5/5] COPY app.py .                                                              0.1s
 => [product-service] exporting to image                                                             3.0s
 => => exporting layers                                                                              1.8s
 => => exporting manifest sha256:d93a19b29e57f64826f985e7661875cd12ce25cf976614844fd0ede2494cc5      0.0s
 => => exporting config sha256:3b23380c751bc00214bfffef3e26c12f00291aa6c69c7271a57fe2f77f47e9fd      0.0s
 => => exporting attestation manifest sha256:967b1fa8f01ddf5b9635beed8f2ab2e250a654805abd6d16d5      0.1s
 => => exporting manifest list sha256:b47dd2e745184481257ae6e2a70b15e78d435d3f0231256c8529c5c4f      0.1s
 => => naming to docker.io/library/week4-lab-product-service:latest                                  0.0s
 => => unpacking to docker.io/library/week4-lab-product-service:latest                               0.9s
 => [product-service] resolving provenance for metadata file                                         0.0s
 => [order-service internal] load build definition from Dockerfile                                   0.1s
 => => transferring dockerfile: 205B                                                                 0.0s
 => [order-service internal] load .dockerignore                                                      0.1s
 => => transferring context: 2B                                                                      0.0s
 => [order-service internal] load build context                                                      0.1s
 => => transferring context: 1.50kB                                                                  0.0s
 => [order-service 3/5] COPY requirements.txt .                                                      0.2s
 => [order-service 4/5] RUN pip install --no-cache-dir -r requirements.txt                          25.4s
 => [order-service 5/5] COPY app.py .                                                                0.2s
 => [order-service] exporting to image                                                               5.3s
 => => exporting layers                                                                              2.8s
 => => exporting manifest sha256:d010fbe36f04416d77ca27e85c972a75853f72f0bb2f2850a61b7bca2fa7a9      0.1s
 => => exporting config sha256:1b5a6760370c8344cb6678f4b9e726b13c09ec9e0e7a2a3f0cb0efecbc054bce      0.1s
 => => exporting attestation manifest sha256:0d063b369ecfcdb614e484e843fc44dc1c3ff3aa7cfbd0b9f6      0.1s
 => => exporting manifest list sha256:1d724b7f093cfa4579508ce769ffa88595759269ea2207055a1007a58      0.1s
 => => naming to docker.io/library/week4-lab-order-service:latest                                    0.0s
 => => unpacking to docker.io/library/week4-lab-order-service:latest                                 2.0s
 => [order-service] resolving provenance for metadata file                                           0.0s
[+] Running 3/3
 ✔Network week4-lab_default  Created                                                                 0.2s
 ✔Container product-service  Started                                                                 1.3s
 ✔Container order-service    Started                                                                 1.5s

C:\Users\DELL\Desktop\week4-lab>docker compose ps
NAME              IMAGE                       COMMAND           SERVICE           CREATED          STA
TUS          PORTS
order-service     week4-lab-order-service     "python app.py"   order-service     18 seconds ago   Up
17 seconds   0.0.0.0:5002->5002/tcp
product-service   week4-lab-product-service   "python app.py"   product-service   18 seconds ago   Up
17 seconds   0.0.0.0:5001->5001/tcp

C:\Users\DELL\Desktop\week4-lab>
```

localhost:5001/health

Pretty-print

{"service":"product-service","status":"up"}

```
C:\Users\DELL\Desktop\week4-lab>curl http://localhost:5001/health
{"service":"product-service","status":"up"}

C:\Users\DELL\Desktop\week4-lab>curl http://localhost:5001/products/1
{"id":1,"name":"Laptop","price":1200}

C:\Users\DELL\Desktop\week4-lab>curl http://localhost:5001/products/99
{"error":"Product not found"}

C:\Users\DELL\Desktop\week4-lab>curl -X POST http://localhost:5002/orders ^
More?  -H "Content-Type: application/json" ^
More?  -d "{\"product_id\":1,\"quantity\":2}"
{"message":"Order created","product":"Laptop","quantity":2,"total_price":2400}

C:\Users\DELL\Desktop\week4-lab>curl -X POST http://localhost:5002/orders ^
More?  -H "Content-Type: application/json" ^
More?  -d "{\"product_id\":1,\"quantity\":2}"
{"message":"Order created","product":"Laptop","quantity":2,"total_price":2400}

C:\Users\DELL\Desktop\week4-lab>
```


localhost:5001/products/1
All Bookmarks
retty-print
"id":1,"name":"Laptop","price":1200}

```
C:\Users\DELL\Desktop\week4-lab>docker stop product-service
product-service

C:\Users\DELL\Desktop\week4-lab>curl -X POST http://localhost:5002/orders ^
More?  -H "Content-Type: application/json" ^
More?  -d "{\"product_id\":2,\"quantity\":1}"
^C
C:\Users\DELL\Desktop\week4-lab>
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C:\Users\DELL\Desktop\week4-lab>
C:\Users\DELL\Desktop\week4-lab>
C:\Users\DELL\Desktop\week4-lab>curl -X POST http://localhost:5002/orders ^
More?  -H "Content-Type: application/json" ^
More?  -d "{\"product_id\":2,\"quantity\":1}"
{"error":"product-service unavailable"}

C:\Users\DELL\Desktop\week4-lab>
C:\Users\DELL\Desktop\week4-lab>
C:\Users\DELL\Desktop\week4-lab>
C:\Users\DELL\Desktop\week4-lab>
C:\Users\DELL\Desktop\week4-lab>1
'1' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\DELL\Desktop\week4-lab>
```